Battery Optimization Analysis for ESP32S3-based Dog Activity Tracker

Battery Specifications

- **Voltage:** 3.7V
- **Capacity:** 350 mAh
- **Energy:** 1.295 Wh

System Overview

- **Platform:** ESP32-S3
- **Interfaces:** BLE (primary for GPS input), Wi-Fi (OTA + data upload)
- **Sensor:** QMI8658 (accelerometer/gyroscope)

Device Behavior

Wi-Fi Usage

- Used only for:
  - OTA firmware updates
  - Periodic upload of logged data
- If offline, all logs are stored locally (SPIFFS/SD).
- Once reconnected to Wi-Fi, stored logs are uploaded.
- Assumed usage: 10 seconds/day @ 70 mA $\Rightarrow$ 0.20 mAh/day

BLE Usage

- Active during motion (e.g., outdoor walks) or on-demand
- Receives GPS coordinates from mobile app
- Assumed usage: 2.5 hours/day @ 0.3 mA $\Rightarrow$ 0.75 mAh/day

Sensor Logging (QMI8658)

- Accelerometer data sampled every 10 seconds
- Active duration per sample: 0.1s
- Average duty cycle: 1%
- Sensor + MCU assumed draw: 10 mA $\times$ 1% $\Rightarrow$ 0.1 mA
- Active for 17 hours/day $\Rightarrow$ 1.7 mAh/day

Sleep Modes

- **Light Sleep:** Triggered after 5–10 minutes of inactivity
  Duration: 5 hours/day @ 0.05 mA $\Rightarrow$ 0.25 mAh
- **Deep Sleep:** After 10+ minutes inactivity or overnight
  Duration: 6 hours/day @ 0.02 mA $\Rightarrow$ 0.12 mAh

Other Assumptions

- **Epilepsy monitoring:** Rare event, 10s/day @ 15 mA $\Rightarrow$ 0.04 mAh/day
- **BLE advertising idle mode:** Negligible power ($<$ 0.01 mAh/day)

Daily Power Consumption Summary

Component	**Avg Current**	**Duration**	**Daily Usage**
Sensor + MCU	0.1 mA	17 h	1.70 mAh
BLE (Connected)	0.3 mA	2.5 h	0.75 mAh
Wi-Fi Burst	70 mA	10 s	0.20 mAh
Light Sleep	0.05 mA	5 h	0.25 mAh
Deep Sleep	0.02 mA	6 h	0.12 mAh
Epilepsy Events	15 mA	10 s	0.04 mAh
BLE Idle	Negligible	¡10 min	0.01 mAh
Total			**3.06 mAh/day**

Battery Life Estimate

$$\text{Battery Life} = \frac{350 \text{ mAh}}{3.06 \text{ mAh/day}} \approx \textbf{114.4 days}$$

Conclusion

- Target battery life (45+ days) is comfortably exceeded.
- Design allows for log upload retries and BLE GPS coordination.
- Future expansion (e.g., more frequent logging) is feasible within current power budget.